

03/30/2026

Quasicoherent sheaves

on affine toric stacks II

Last time started the proof that $\mathrm{QCoh}(X_\sigma/T) \simeq \mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp})$,
the universal property of Day Convolution.

Today Step 1 of the proof the Fourier transform, (co)monadic descent.

Recall For a lattice N and cone $\sigma \subset N_{\mathbb{R}}$:

(1) We write

$$\mathrm{Mod}_{S_\sigma}(An)^M := \lim_{[n] \in \Delta} \mathrm{Mod}_{S_\sigma \times M^{\times n}}(An),$$

where $[n] \mapsto S_\sigma \times M^{\times n}$ is the diagram defining the coaction of S_σ on M .

(2) Taking $S \langle - \rangle$ termwise and taking limits, we obtain a
Sym mon functor

$$\Theta[-] : \mathrm{Mod}_{S_\sigma}(An)^M \longrightarrow \mathrm{QCoh}(X_\sigma/T).$$

(3) We explicitly defined a symmetric monoidal functor

$$\phi_\sigma : \Theta(\sigma) \longrightarrow \mathrm{Mod}_{S_\sigma}(\mathrm{Set})^M.$$

So we have functors

$$\Phi_\sigma: \Theta(\sigma) \xrightarrow{\phi_\sigma} \text{Mod}_{S_\sigma}(\text{Set})^M \xrightarrow{\quad} \text{Mod}_{S_\sigma}(\text{An})^M \xrightarrow{\mathcal{O}[-]} \text{QCoh}(X_\sigma/\tau)$$

Sym. mon
 Not Sym. mon
Sym. mon

Question So it seems we have a problem. How do we resolve this?

Symmetric monoidality of Φ_σ

Observation. Let $S \in \mathbf{CMon}(\mathbf{Set})$. Let $\text{Mod}_S^{\text{free}}(\mathbf{An}) \subset \text{Mod}_S(\mathbf{An})$ denote the full subcategory spanned by coproducts of S .

- > Then $\text{Mod}_S^{\text{free}}(\mathbf{An}) \subset \text{Mod}_S(\mathbf{An})$ is closed under the relative tensor product $\otimes_S$.
- > Moreover, $\text{Mod}_S^{\text{free}}(\mathbf{An})$ coincides with $\text{Mod}_S^{\text{free}}(\mathbf{Set}) \subset \text{Mod}_S(\mathbf{Set})$.

Observation. For a lattice N and cone $\sigma \in N_{\mathbb{R}}$, write

$$\text{Mod}_{S_\sigma}^{\text{free}}(\mathbf{Set})^M := \lim_{[n] \in \Delta} \text{Mod}_{S_\sigma \times M \times n}^{\text{free}}(\mathbf{Set})$$

That is, $\text{Mod}_{S_\sigma}^{\text{free}}(\mathbf{Set})^M \subset \text{Mod}_{S_\sigma}(\mathbf{An})^M$ is the full subcategory spanned by those objects such that each term is free.

- > Then the inclusion $\text{Mod}_{S_\sigma}^{\text{free}}(\mathbf{Set})^M \hookrightarrow \text{Mod}_{S_\sigma}(\mathbf{An})^M$ is symmetric monoidal.

Observation. For each $V \in \Theta(\sigma)$, we have $V = m + \sigma^V$ for some $m \in M$. So

$$V \cap M = m + (\sigma^V \cap M) = m + S_\sigma.$$

As an S_σ -module, this is isomorphic to S_σ , hence free.

Upshot. The functor

$$\Theta(\sigma) \xrightarrow{\phi_\sigma} \text{Mod}_{S_\sigma}^{\text{free}}(\text{Set})^M \hookrightarrow \text{Mod}_{S_\sigma}(\text{An})^M \xrightarrow{\mathcal{Q}[-]} \text{QCoh}(X_\sigma/T)$$

is sym mon. Hence this induces a sym. mon. left adjoint

$$\Phi_\sigma : (\text{Fun}(\Theta(\sigma)^{\text{op}}, \text{Sp}), \otimes_{\text{Day}}) \longrightarrow \text{QCoh}(X_\sigma/T).$$

Exercise. Let Σ be a fan in N . Check that the functors ϕ_σ assemble into a natural transformation

$$\Theta(-) \longrightarrow \text{Mod}_{S_{(-)}}(\text{An})^M$$

of functors $\Sigma^{\text{op}} \rightarrow \text{CMon}(\text{Cat}_\infty)$.

• As a result, the functors Φ_r assemble into a natural transformation

$$\Theta(-) \longrightarrow \mathrm{QCoh}(X_G/\mathbb{T})$$

of functors $\Sigma^{\mathrm{op}} \rightarrow \mathrm{CMon}(\mathrm{Cat}_{\infty})$.

Goal. Show that each Φ_r is an equivalence.

The Fourier transform

First we'll try to understand quasicoherent sheaves on the classifying stack

$$BT = */T$$

> understanding how both sides map to $QCoh(BT)$ will basically allow us to reduce to this situation

Notation. For $M \in \text{Grp}(\text{Set})$ an abelian group, write $D_M = \text{Spec}(\mathbb{S}\langle M \rangle)$ for the corresponding commutative group in Stk

Notn. For $M \in \text{Grp}(\text{Set})$, consider the trivial group 0 as a comodule over M . Write

$$\text{Mod}_0(A_n)^M = \lim_{[n] \in \Delta} \text{Mod}_{0 \times M \times n}(A_n).$$

Construction. For $M \in \text{Grp}(\text{Set})$, write

$$\phi_M: M \longrightarrow \text{Mod}_0(\text{Set})^M$$

for the functor

$$m \longmapsto (0, f_m: 0 \times M \longrightarrow 0 \times M)$$

where $f_m: M \longrightarrow M$ is $n \mapsto m+n$.

Check (easy). This satisfies the cocycle condition.

This assignment clearly admits a natural sym mon structure.

> Like before, ϕ_M factors through $\text{Mod}_0^{\text{free}}(\text{Set})^M$ and the inclusion

$$\text{Mod}_0^{\text{free}}(\text{Set})^M \subset \text{Mod}_0(\text{An})^M$$

is symmetric monoidal.

Note We also have a natural sym mon. functor

$$\text{Mod}_0(A_n)^M = \lim_{[n] \in \Delta} \text{Mod}_{0 \times M^{x_n}}(A_n) \xrightarrow{\mathcal{Q}[-]} \lim_{[n] \in \Delta} \text{QCoh}(* \times D_M^{x_n}) \simeq \text{QCoh}(\text{BD}_M)$$

induced by taking $S \mapsto$ termwise

Upshot We have a sym mon functor

$$M \xrightarrow{\Phi_M} \text{Mod}_0^{\text{free}}(\text{Set})^M \hookrightarrow \text{Mod}_0(A_n)^M \xrightarrow{\mathcal{Q}[-]} \text{QCoh}(\text{BD}_M)$$

we write Φ_M for the induced sym mon. left adjoint

$$\Phi_M(\text{Fun}(M, \text{Sp}), \otimes_{\text{Day}}) \longrightarrow \text{QCoh}(\text{BD}_M)$$

Thm (Fourier transform) For $M \in \text{Grp}(\text{Set})$, the sym mon. functor

$$\Phi_M(\text{Fun}(M, \text{Sp}), \otimes_{\text{Day}}) \longrightarrow \text{QCoh}(\text{BD}_M)$$

is an equivalence

- > To prove this, we'll need to introduce a useful categorical tool.
- > Before doing so, we make an important observation.

Observation (Compatibility with Φ_σ) Let N be a lattice, $\sigma \in N_{\mathbb{R}}$ a cone, and $M = N^\vee$ the dual lattice. Write $p_\sigma: M \rightarrow \Theta(\sigma)$ for the Sym. mon. functor $m \mapsto m + \sigma^\vee$. (Again, we use that $\sigma^\vee + \sigma^\vee = \sigma^\vee$.)

- > Write $\pi_\sigma: X_\sigma / T \rightarrow * / T = BT$ for the induced map of quotient stacks.

We claim that the diagram

$$\begin{array}{ccc}
 \text{Fun}(M, S_p) & \xrightarrow{\Phi_M} & \text{QCoh}(BT) \\
 \text{left Kan} & \downarrow p_{\sigma, \#} & \downarrow \pi_\sigma^* \\
 \text{extension} & & \\
 \text{along } p_\sigma & \text{Fun}(\Theta(\sigma)^\sigma, S_p) & \xrightarrow{\Phi_\sigma} \text{QCoh}(X_\sigma / T)
 \end{array}$$

canonically commutes. Indeed, by the universal property

of Day convolution, it suffices to check that the restriction to M along the spectral Yoneda embedding commutes. This reduces to the claim that the square of 1-categories

$$\begin{array}{ccc}
 M & \xrightarrow{\phi_M} & \text{Mod}_0(\text{Set})^M \\
 \downarrow p_\sigma & & \downarrow \text{induced by } 0 \rightarrow S_\sigma \text{ as } M\text{-comodules} \\
 \Theta(\sigma) & \xrightarrow{\phi_\sigma} & \text{Mod}_{S_\sigma}(\text{Set})^M
 \end{array}$$

commutes, which is clear from unpacking the definitions

Tool: (Co)monadic descent

The situation we're in. We have a functor

$$\Phi_M: \text{Fun}(M, \text{Sp}) \longrightarrow \text{QCoh}(\text{BD}_M) = \lim_{[n] \in \Delta} \text{Mod}_{S\langle M^{\times n} \rangle}$$

that we want to show is an equivalence

> We'll abstract the setup.

Ntn. Let $\Delta_+ \simeq (\Delta)^{\Delta}$ denote the category of (possibly empty) finite linearly ordered sets and order-preserving maps

Write $[-1] = \emptyset$

> A diagram indexed by Δ_+ is called an augmented cosimplicial diagram

Q Given an augmented cosimplicial ∞ -category $\mathcal{C}^{\bullet}: \Delta_+ \rightarrow \text{Cat}_{\infty}$, What are some checkable conditions that guarantee that

$$\mathcal{C}^{-1} \xrightarrow{\sim} \lim_{[n] \in \Delta} \mathcal{C}^n$$

> The answer comes from the theory of (co)monads and their ∞ -categories of (co)modules.

- But to formulate the answer, we'll need less.

Recollection (Split equalizers) Let $\mathcal{C}$ be a 1-category. A split equalizer diagram in $\mathcal{C}$ is a diagram

$$\begin{array}{ccccc} & & \xleftarrow{h^{-1}} & & \\ & & & & \\ X^{-1} & \xrightarrow{d^0} & X^0 & \xrightarrow{d^0} & X^1 \\ & & & & \\ & & \xleftarrow{h^0} & & \\ & & & & \end{array}$$

d^1

Satisfying

$$d^0 d^0 = d^1 d^0, \quad h^{-1} d^0 = \text{id}_{X^{-1}}$$

$$h^0 d^0 = \text{id}_{X^0}$$

$$h^0 d^1 = d^0 h^{-1}$$

} version of
Simplicial
identities
including -1

Lem. Every split equalizer diagram is an equalizer, hence all functors preserve these equalizers.

> As usual, we want a version for simplicial objects

Def. Write $\Delta_{\min} \subset \Delta$ for the non-full subcategory with all objects but only those maps that preserve minimal elements i.e., $\alpha: [m] \rightarrow [n]$ such that $\alpha(0) = 0$.

Observation. The inclusion $\Delta_{\min} \subset \Delta_+$ admits a left adjoint $C_+: \Delta_+ \rightarrow \Delta_{\min}$ given by $I \mapsto [0] * I$ so $[n] \mapsto [n+1]$.

Def. Let $X^\bullet: \Delta_+ \rightarrow \mathcal{C}$ be an augmented cosimplicial object. A splitting of $X^\bullet$ is a factorization

$$X^\bullet: \Delta_+ \xrightarrow{C_+} \Delta_{\min} \xrightarrow{\bar{X}^\bullet} \mathcal{C}$$

We say that $X^\bullet$ is split if $X^\bullet$ admits a splitting.

Observation (extra codegeneracies). For $n \geq -1$, in $\Delta_{\min}$ we have the map $\sigma^n: [n+2] \rightarrow [n+1]$ given by

$$i \mapsto \begin{cases} 0, & i=0 \\ i-1, & i>0 \end{cases}$$

So a splitting $\bar{X}^\bullet$ of $X^\bullet$ provides extra maps

$$h^n: X^{n+1} \simeq \bar{X}^{n+2} \xrightarrow{\bar{X}^\bullet(\sigma^n)} \bar{X}^{n+1} \simeq X^n$$

These maps come equipped with equivalences

$$d^i \circ h^n \simeq \begin{cases} \text{id}_{X^n}, & i=0 \\ h^{n-1} \circ d^{i-1}, & \text{else} \end{cases} \quad \text{and} \quad s^i \circ h^n \simeq \begin{cases} h^{n+1} \circ h^n, & i=0 \\ h^{n+1} \circ s^{i-1}, & \text{else} \end{cases}$$

The maps $(h^n)_{n \geq -1}$ are called extra degeneracies

Exercise If $\mathcal{C}$ is a 1-category, show that there is a bijection between the set of splittings of $X^\bullet$ and the set of collections of maps $(h^n)_{n \geq -1}$ satisfying the above conditions.

Lem. If $X^\bullet: \Delta_+ \rightarrow \mathcal{C}$ splits, then $X^\bullet$ is a limit diagram.

> we need one more definition to state the result we'll need.

Def. Let

$$(*) \quad \begin{array}{ccc} A & \xrightarrow{\bar{f}^*} & B \\ \bar{g}^* \downarrow & \nearrow \sigma & \downarrow g^* \\ C & \xrightarrow{f^*} & D \end{array}$$

be a square of ∞ -categories commuting up to a specified natural trans $\sigma: f^* \bar{g}^* \rightarrow g^* \bar{f}^*$

Assume that f^* and $\bar{f}^*$ admit right adjoints f_* and $\bar{f}_*$.

The exchange transformation is the composite

$$\text{Ex: } \bar{g}^* \bar{f}_* \xrightarrow{\text{unit}} f_* f^* \bar{g}^* \bar{f}_* \xrightarrow{\sigma} f_* g^* \bar{f}^* \bar{f}_* \xrightarrow{\text{counit}} f_* g^*$$

We say that $(*)$ is horizontally right adjointable if f^* and $\bar{f}^*$ admit right adjoints and Ex is an equivalence.

Thm (Beck for 1-categories, Lurie for ∞ -categories) Let $e^*: \Delta_+ \rightarrow \text{Cat}_\infty$ be an augmented cosimplicial ∞ -category write $G: e^{-1} \rightarrow e^0$ for the augmentation Assume

(1) G preserves limits of cosimplicial diagrams that become split after applying G .

(2) For each map $\alpha: [m] \rightarrow [n]$ in Δ_+ , the commutative square

$$\begin{array}{ccc}
 e^m & \xrightarrow{d^0} & e^{m+1} \\
 \alpha_* \downarrow & \curvearrowright R_m & \downarrow ([0] + \alpha)_* \\
 e^n & \xrightarrow{d^0} & e^{n+1} \\
 & \curvearrowright R_n &
 \end{array}$$

is horizontally right adjointable.

$$\alpha_* R_m \xrightarrow{\sim} R_n \circ ([0] + \alpha)_*$$

Then $e^{-1} \rightarrow \lim_{[n] \in \Delta} e^n$ admits a fully faithful right adjoint.

If G is also conservative, then this functor is an equivalence.